

Engineering optimization of LLMs: some consequences for debates in cognitive science

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Abstract

We highlight a meta-optimization process of LLMs, which we refer to as R&D. We show that this technological process matters when modern language systems are used to advance important debates in AI and cognitive science. In particular, the successes of LLMs currently do not inform us sufficiently about human language learning biases.

Keywords: *large language models, optimization, natural selection, learning biases*

1. Introduction

There has been a recent explosion of work on questions pertaining to whether large language models (LLMs) display human-level understanding (Bender & Koller, 2020; Ohmer et al., 2024), have access to linguistic structure or semantic meaning (Baggio & Murphy, 2024; Mandelkern & Linzen, 2024; Moro et al., 2023) or even possess a ‘theory of mind’ (Hu et al., 2025; Strachan et al., 2024). At the heart of the matter is the issue of learning, i.e. what kind of learners are LLMs, and how does this compare to the kind of learning exhibited in humans (e.g., Lan et al., 2024).

To contribute to this debate, we focus on a specific kind of optimization process, a sort of technological evolution (Bickel et al., 2024), which has been massively important in the recent rise of deep learning systems in AI. We show how these processes make it harder to sustain arguments of the following form: LLMs master Capacity X, therefore there is no specific learning bias needed to master Capacity X, or therefore the biases used by humans to master Capacity X are such and such, or therefore LLMs are good theories of human learning. Such arguments would be fundamental in various cognitive science debates. We exhibit places where it is tempting to think that we now have the resources to address these issues thanks to what we see in LLMs, but we call for caution.

2. Two Kinds of Optimization: R&D and TRAINING

LLMs are engineering devices designed for specific tasks, in this case, production and prediction of human language text. They are input/output functions, in the form of neural networks, which are parametrized by “weights”. LLMs achieve their success through an

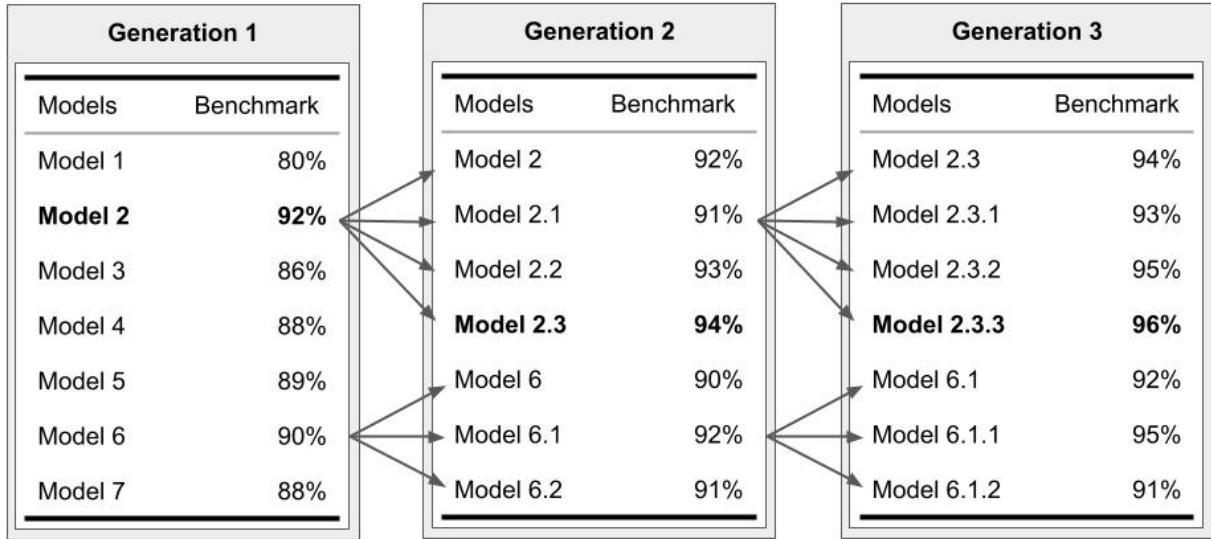


Figure 1: LLMs have followed a long chain of optimization, which have made them increasingly specialized for language learning.

optimization phase called TRAINING, during which the weights are gradually adjusted to best fit the desired input/output relation.

But not all LLMs are equal. LLMs may go through different types of training regimes, governed by what are called hyperparameters (and possibly also meta-hyperparameters, which could govern how the space of hyperparameters is itself searched). LLMs may also be based on different “architectures”, defined even before they enter TRAINING: weights may be organized differently to combine into functions of different forms. Hence, any LLM is the result of choices concerning which architecture to use and which training regime to deploy. These choices are informed by a process that we call R&D: during R&D, multiple options have been tried and tested for e.g. architectures, and we gradually learn which options work best.

R&D can be seen as an optimization process, happening before the more visible optimization process, TRAINING. And one can trace its history. In short, connectionism established the cognitive scientific relevance of recurrent neural networks for language processing (Elman, 1991). For a long time, RNNs were the preferred tools of language modelling in NLP as a result. In this light, the emergence of the transformer architecture (Vaswani et al., 2017) could be seen as a form of macromutation.¹ A small change from recurrence as a mechanism to self-attention massively influenced future model architecture, and became the default option for the design of LLMs. R&D optimization happened through such macro-changes as well as micro-changes, and quite explicitly. In academia, the standard format of results in the domain is a plain table of numbers, whereby models (in rows) are compared according to established benchmarks (in columns). For each benchmark, the model that surpasses the others is traditionally highlighted with boldface, and it is the one that receives most attention from the community, and attracts follow-up studies, see Fig. 1. In other words, models are evaluated and compared based on some (manually crafted) fitness measures, and those that strive get to be improved further in the next generations of studies.

Hence, the models that we know today—BERT, Llama, Gemini, GPT-3, 4, 5, LaMDA—

¹In fact, Oh & Linzen (2025) argue that LLMs have become ‘superhuman’ and thus less relevant to cognitive science. As evidence they cite the fact that the earlier, more connectionist, language models aligned more closely with human reading times. However, around 2022, fuelled in part by the rise of transformer architectures, this course dramatically reversed.

are not default or random samples of a diverse population of models; they are the models selected for their performance, further optimized to perform even better, while others have been selected out for their lack of relative accuracy.²

3. The extent of R&D optimization

We presented R&D as a full-blown search-optimization process, a pre-optimization process compared to TRAINING. Optimization happens through small and large changes (e.g., small changes in the number of layers of a model, or more significant innovations like the move to transformer-based architectures). It is driven by a specific target (the modern form of human language) and explicit engineering decisions. These are very strong advantages for an optimization process, compared to a process of evolution trying to find *some* solution to a general fitness problem without assumptions about what this solution would look like in advance. And here we show that the magnitude of the process is also strong.

Let us constrain ourselves to the deep learning NLP paradigm which emerged in the 2010s. Let us assume that an initial conservative spread of around 1,000 labs tested an average of 10 LLMs per year. This would mean 1×10^4 different LLMs within a 12 month period. If the number of labs increased (with funding) at a constant rate of 50% year over year, then by 2024 we already have close to 10^8 tested LLMs. And if these astronomic numbers seem exaggerated, we should add that there is no reason to think that the growth rate has been constant (a common estimate is that the investment effort in AI grew at 80% in 2024) nor that every lab yields an equal number of tested LLMs with major players such as OpenAI, Google DeepMind, and Meta likely significantly adding to these numbers. Nevertheless, our estimate merely indicates the possible space of variation within LLM generation over a 15 year period. In comparison, a regularly cited estimate from the Population Reference Bureau of the number of homo sapiens sapiens who ever lived is of a similar order of magnitude at 10^9 people.³

So, LLMs have followed an optimization process which we call R&D. If a comparison was to be made, this process is possibly a much stronger and more efficient one than nature provides, given the extent of the search, and the explicit engineering forces involved to specifically get at human language in its current form (an outcome that was not available as an explicit target to nature *a priori*). The fact that any LLM born today has followed such an optimization process, above and beyond its own TRAINING on actual data, has massive consequences for how we may use their success to illuminate debate in cognitive science in general and human language learning in particular. We discuss this in the next section.

4. Comparing humans and LLM learning biases

While this is well-known, let us reiterate clearly that all learners have biases. Suppose that a learner is exposed to a finite dataset. It may assume that what it has been exposed

²Yax et al. (2024) can be seen as pushing the natural evolution metaphor, as they propose a way to construct phylogenetic trees of LLMs (primarily in terms of ‘traits’, that is their behavior as to what tokens they generate and how they fare with respect to various benchmarks, rather than direct structural or historical considerations). Surely, human/animal evolution is a different process, eg, because natural evolution is a bottom-up blind process. We refer however to Dennett (2017) for an evolutionary account of top-down intelligent systems like LLMs.

³<https://www.prb.org/international/>

to is everything there is. Or it may assume that, given the diversity of what it has seen, everything is possible. Those options are two possible types of biases. And without biases there is no learning (among many others, see Rawski & Heinz, 2019).

Humans thus come to the task of learning their native language with learning biases. Two questions for LLMs then arise. First, are the biases of modern LLMs similar to those of humans? Second, if so, does it tell us that human biases are somewhat unexceptional? In this section, we address the former question (and move to the latter in the next section). What can be used to say that biases of LLMs and humans are the same?

Strong outcomes. The most attractive answer is based on outcomes: given that, in a variety of tasks, LLMs approximate outcomes of humans very well (learning went well), this must come about through similar learning biases. Surely, LLMs outputs are very human-like. But without equating the TRAINING conditions, one cannot say whether the biases are the same. And TRAINING in humans and learning surely is not comparable, if only in magnitude (Frank, 2023), or in its multimodality aspect. Furthermore, approximating the outcome is not enough either. Suppose that a system approximates another with a precision of 99%. This is certainly very powerful for technological purposes, but from a scientific purposes it is crucial to inspect the last 1%. There is a goal discrepancy here, unsurprising between engineering and science (here, linguistics): engineering is interested in covering as many cases as possible, so typically the most frequent cases have a strong weight, while science may insist on capturing the tail of the distribution because even if it may not have any real world impact, it is the best potential window into which generalizations are real. Because these phenomena are rare at training, they may also be where the learning biases alone express themselves the best.

Impossible languages. A second set of empirical comparison goes beyond mere output, and asks whether LLMs learn actual languages better than distorted variants, so-called “impossible languages” (Mitchell & Bowers, 2020; Moro et al., 2023). An impossible language is constructed as a set of strings with properties of order or configuration that is not possible for a human to acquire during the language acquisition process, arguably because no attested human language has this property. Moro (2016) exemplifies this notion with the case of linearly ordered languages, as opposed to hierarchically embedded as in a recursive phrase-structure grammar, but there are other possible properties to be considered.

The empirical situation here is complex, and a short summary could go as follows: Moro and others argue that there is no principled reason why, say, order biases would be implemented in the architecture of LLMs and predicts that some humanly ‘impossible’ language may look perfectly plausible to LLMs (e.g., Moro, 2016); some results offer signs of preferences for possible languages nonetheless by LLMs (Kallini et al., 2024); further investigations provide very mixed results when more impossible languages are taken into account, or even simply more reference languages beyond English (Ziv et al., in press).

Learning stages. Humans seem to be learning their native language in phases that have been more and more systematically described (e.g., Friedmann et al., 2021) and, simply put, the early phases may be those aligning best with the strongest learning biases. A recent line of investigation has looked in more detail at the different stages of learning in LLMs, asking whether the same learning phases would emerge (Evanson et al., 2023). This is a very ambitious literature, which ought to provide powerful references to compare learning biases from a very fine angle. But it is too early to conclude from it that there are similar learning biases in humans and LLMs.

Machine generated prompts. Finally, another relevant set of facts in comparing learner biases comes from looking at what can be described anthropomorphically as overgeneralizations by LLMs. It has been shown in the prompt engineering literature that, if LLMs respond well to English, they also seem to respond as well if not better to a secret language that they generated alongside English: cryptic sequences of symbols seem to let them answer some questions more efficiently than plain English (Deng et al., 2022; Kervadec et al., 2023; Shin et al., 2020; Wallace et al., 2019; Zhong et al., 2021). What this shows is that their training leads them to ‘generalize’ in ways that we would not generalize. While this is convenient from an engineering perspective, and while it does reveal rich learning biases in current LLMs, we take this to be an underrated argument that the biases of current LLMs are not like ours.

5. Optimizations and debates in cognitive science

For a moment, let us assume that the achievements of LLMs are such that they could raise the possibility that their biases are similar to that of humans. This may currently seem like an enormous assumption, but the past decade of progress had made vivid the possibility that we may someday manage to develop such an impressive Frankenstein model. Our main point is to reflect on the exact consequence of the existence of such a model for our understanding of human biases.

There is a temptation to believe that if a machine models our biases, then there would be nothing impressive in our biases. But now we need to remember R&D: these machines have been optimized to fit these biases. Thus, the fact that they end up fitting them, might tell us we have succeeded in the approximation goal, but it still tells us very little about the biases themselves. For example, this does not tell us if these biases are strong or weak, simple or complex, if learning a language is a difficult or simple learning task (assuming that these terms could be defined in the first place). If any machine one could have thought about could learn language, then the situation would be different. At this stage, progress crucially requires that we better understand the biases of the successful models. And despite progress in this area, we still do not have a complete understanding of these biases.

We may also ask what we learn about modularity (see discussion in Fox & Katzir, 2024), or here a possible specificity of language learning biases. There again it is hard to conclude: On the one hand the machines have been optimized for learning languages specifically, and on the other hand the literature across different domains is permeable (which is for the best: architectures develop for say, vision, may be tested with language, and vice-versa). We may in the future build machines showing human-like learning, learning language and other behaviors. But depending on how they would have been optimized, how much we understand of how these machines work (as opposed to just being able to build them), and, to our crucial point, how they have been pre-optimized by R&D, it may or may not be informative on the human question of learning biases, their history and their specificity.

6. R&D and TRAINING vs. evolution and learning?

Simplifying, one could see R&D as a specialization of LLMs for human language, in the same way that evolution has carved specific biases for humans. One important difference, as discussed above, is that evolution is blind to the final objective and could have ended

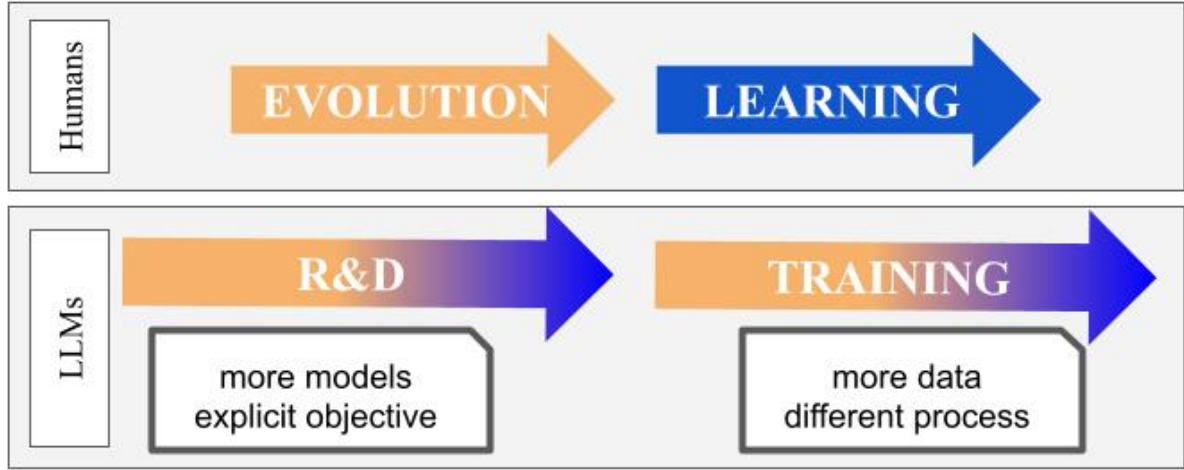


Figure 2: The two phases of optimizations in humans and in LLMs have different scopes, different properties, and do not necessarily align well.

up finding a different solution for survival, while R&D works with the narrower objective of matching the solution that evolution has endowed.

And the alignment between R&D and TRAINING on the one hand for LLMs, and evolution and learning on the other hand for humans, is in fact not straightforward (cf. Fig. 2). Some parts of the optimization that was in one bucket for humans could be in another bucket for LLMs. So, R&D has happened under a background such that TRAINING is done with a lot of data, and maybe this extra data makes it unnecessary for the biases to be optimized in the same way at R&D time. But then what in LLM development can be taken as a model of the human learning phase becomes blurry.

If LLMs were to be taken as a model of human language learning, there would therefore be a blurry division of labour between where evolution and biases are modelled and where learning is. Some are interested in the former: what biases did evolution equip humans with? Some are interested in the latter: how does learning proceed? In both cases, the success of LLMs are hard to interpret given what we have highlighted about R&D, and a misalignment between the optimization stages in humans and in machines.

7. Conclusions

LLMs mark a remarkable human achievement. However, the role they play in the theory of language learning and cognitive science in general is still much contested. Part of the dispute concerns whether their successes are sufficient to claim them human-like behavior in the first place, or whether their training regime is comparable at all to human learning. Even if we granted perfect human-behavior and comparable learning, now or in the future, we argue that the way their engineering successes weigh in fundamental debates related to linguistic competence, language acquisition, and the limits of human cognition must be assessed with care, and it must take into account their history and engineering trajectory, or R&D. These include optimization processes, of which we still do not fully understand the consequences, and in particular what biases they inject.

LLMs are learners, learners with biases, as are all learners. Their biases have been and continue to be sculpted efficiently. Prior to any TRAINING, they followed a meta-optimization process led by both insightful and/or serendipitous innovations, benefiting from bottom-up as well as top-down signals, that are injected explicitly or implicitly into them. Humans are learners, learners with biases as are all learners. Prior to birth, they

followed a meta-optimization process led by nature. 225

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